

Below, $\mathbb{1}()$ will denote the indicator or truth function. $\mathbb{1}(S)$ equal 1 if S is true, and 0 if S is false. For example, $\mathbb{1}(x \geq 0)$ equals 0 if $x < 0$ and 1 otherwise.

Standardized Precipitation Index (SPI) ranges from -4 (least rain) to 4 (most rain).

$I_D = \mathbf{Impact}$ (i.e., **severity**) of drought is defined as minus SPI (higher rain, higher SPI, less severe drought)

$P(I_D)$ = likelihood, probability that a drought of severity I_D will occur in the near future (weeks to months), based on forecasts.

μ = ensemble member index

M = number of ensemble members, 51 for *ECMWF* (European Centre for Medium-Range Weather Forecasts)

$$Risk_D = E[\underline{I}_D] = \frac{1}{M} \sum_{\mu=1}^M I_{D,\mu} \quad (1)$$

$$= \sum_{I_D} P(I_D) I_D \quad (2)$$

A **risk matrix** is a 4×4 matrix with X axis equal to impact (I) and Y axis equal to likelihood P , with I binned into four values and $P(I)$ binned into 4 values. The matrix elements equal to some arbitrary function $f(I, P)$ defined as risk. Usually, $f(I, P) = IP$ but not necessarily. I find risk matrices arbitrary and not very useful. The Wikipedia article on risk matrices (Ref.[1]) agrees.

I_D^* = impact threshold for drought (e.g., $I_D^* = -SPI = 1.5$)

$$P(\underline{I}_D \geq I_D^*) = \frac{1}{M} \sum_{\mu=1}^M \mathbb{1}(I_{D,\mu} \geq I_D^*) \quad (3)$$

P_D^* = probability threshold for drought

C_D = cost of drought

L_D = loss of drought

$$P_D^* = \frac{C_D}{L_D} \quad (4)$$

T_D = trigger for drought

$$T_D = \mathbb{1}[P(\underline{I}_D \geq I_D^*) > P_D^*] \quad (5)$$

Let N^D be the number of times (seasons) in which predicted and observed droughts were compared.

$N_{y|x}^D$ = number of events in which $x \in \{0, 1\}$ droughts occurred and $y \in \{0, 1\}$ droughts were predicted

For example, $N_{1|0}^D$ is the number of events in which no drought actually occurred but it was predicted nonetheless (False Alarm)

for clarity, will omit D superscript from N^D below

$$N_{|0} = N_{0|0} + N_{1|0} \quad (6)$$

$$N_{|1} = N_{0|1} + N_{1|1} \quad (7)$$

$$N = N_{|0} + N_{|1} \quad (8)$$

$$\mathbf{Hit\ Rate\ (HR)} = P(1|1) = \frac{N_{1|1}}{N_{|1}} \quad (9)$$

$$\mathbf{False\ Alarm\ Rate\ (FAR)} = P(1|0) = \frac{N_{1|0}}{N_{|0}} \quad (10)$$

$$\mathbf{Miss\ Rate\ (MR)} = P(0|1) = \frac{N_{0|1}}{N_{|1}} \quad (11)$$

$$\mathbf{Specificity} = P(0|0) = \frac{N_{0|0}}{N_{|0}} \quad (12)$$

References

- [1] Wikipedia. Risk matrix. https://en.wikipedia.org/wiki/Risk_matrix.